Mean of the random variable X

$$= E(X) = \sum_x x\,P_X(x)$$

$$= \left(1\times\frac{1}{6}\right)+\left(2\times\frac{1}{6}\right)+\left(3\times\frac{1}{6}\right)+\left(4\times\frac{1}{6}\right)+\left(5\times\frac{1}{6}\right)+\left(6\times\frac{1}{6}\right)$$

$$= \frac{1}{6}\left(1+2+3+4+5+6\right)$$

$$= \frac{7}{2}$$

$$E\left(X^2\right) = \sum_x x^2\,P_X(x)$$

$$= \left(1^2\times\frac{1}{6}\right)+\left(2^2\times\frac{1}{6}\right)+\left(3^2\times\frac{1}{6}\right)+\left(4^2\times\frac{1}{6}\right)+\left(5^2\times\frac{1}{6}\right)+\left(6^2\times\frac{1}{6}\right)$$

$$= \frac{1}{6}\left(1^2+2^2+3^2+4^2+5^2+6^2\right)$$

$$= \frac{91}{6}$$

Variance of the Random Variable

$$X = V(X) = E\left(X^2\right)-\left[E(X)\right]^2$$

$$= \frac{91}{6}-\left(\frac{7}{2}\right)^2$$

$$= \frac{35}{12}$$

Example 6.15

The following information is the probability distribution of successes.

No. of Successes	0	1	2
Probability	$\frac{6}{11}$	$\frac{9}{22}$	$\frac{1}{22}$

Determine the expected number of success.

Solution:

Expected number of success is

$$E(X) = \sum_x x\,P_X(x)$$

$$= \left(0\times\frac{6}{11}\right)+\left(1\times\frac{9}{22}\right)+\left(2\times\frac{1}{22}\right)$$

$$= \frac{11}{22} = 0.5$$

Therefore, the expected number of success is 0.5. Approximately one success.

Example 6.16

An urn contains four balls of red, black, green and blue colours. There is an equal probability of getting any coloured ball. What is the expected value of getting a blue ball out of 30 experiments with replacement?

Solution:

Probability of getting a blue ball (p) = $\frac{1}{4}$

$= 0.25$

Total experiments (N) $= 30$

Expected value = Number of experiments × Probability

$= N \times p$

$= 30 \times 0.25$

$= 7.50$

Therefore, the expected value of getting blue ball is approximately 8.

Example 6.17

A fair die is thrown. Find out the expected value of its outcomes.

Solution:

If the random variable X is the top face of a tossed, fair, six sided die, then the probability mass function of X is

$$P_X(x) = \frac{1}{6},\ \text{for } x = 1,2,3,4,5 \text{ and } 6$$

The average toss, that is, the expected value of X is

$$E(X) = \sum_x x\,P_X(x)$$

$$= \left(1\times\frac{1}{6}\right)+\left(2\times\frac{1}{6}\right)+\left(3\times\frac{1}{6}\right)+\left(4\times\frac{1}{6}\right)$$

$$+\left(5\times\frac{1}{6}\right)+\left(6\times\frac{1}{6}\right)$$

$$= \frac{1}{6}\left(1+2+3+4+5+6\right) = \frac{7}{2} = 3.5$$

XII Std - Business Maths & Stat EM Chapter 6.indd 137 2/28/2020 2:56:13 PM